

» Lessons On Continuous Business Model Innovation With Ash Maurya

Ash Maurya is a practitioner and entrepreneur. Author of Running Lean, Scaling Lean and the Lean Canvas, built on top of the Business Model Canvas, Ash is also the founder of LEANSTACK.

In this session, we focused on continuous innovation and scaling lean.

How did we go from lean manufacturing to continuous innovation?

Ash Maurya: If we go back to, let's say, the last hundred years or so, that's when we were in that manufacturing era, and the unfair advantage all companies were all about mass production.

The companies that produced the most amount of products for the lowest costs tended to win. It was all about efficiency. After the war, a number of companies started to get squeezed, and this is where Taiichi Ohno over at Toyota invented the Toyota Production System, which kind of spawned this new way of thinking. Taking a lot of Just in Time techniques and bringing in what became eventually Lean Manufacturing. That's kind of the origin that we trace back a lot of even what we talk about today in the startup world.

Some of those core principles go back to this idea of being less wasteful, and continuous improvement. Now it has morphed over the years, so as the world has changed, **as we have moved from that manufacturing era to more digital products, the need for speed became ever more important.**

As we got into PC computing, requirements began to change faster and faster, and so methodologies and frameworks evolved. **We moved from traditional manufacturing to waterfall.**

Some of you may have lived through that. We then moved from waterfall to agile where we began to bring in more iterations, and feedback along the way to where we are today.

When Lean Startup came on the scene, the big shift then was our move away from even just PCs to the internet.

As we moved onto the internet, the connection between us and our customers almost vanished. **We're more connected today than ever before to customers, which means that we can learn faster, but also it means that customers demand more than ever before.**

Our need to go faster has reached a certain point to where we as a company to LEANSTACK started calling this continuous innovation. **It's no longer stop and go, we can't take six months to do requirements analysis and then another six months to build a product.**

If you look at some of the top companies today, the likes of Amazon, Facebook, Netflix, all of these companies are continuously building and launching products. The stages between requirements gathering and building and testing have all blurred together. It's all happening continuously.

Gennaro Cuofano: Let me touch another critical point as today there is a sort of misconception that as we start using more and more, the scientific method for entrepreneurs.

Many people get confused between science and entrepreneurship when they're two separate domains.

What's the key difference between an entrepreneur and a scientist?

Ash Maurya: the difference is in goals. Certainly, the analogy helps. A lot of entrepreneurs tend to fly by guts. Gut feeling and instinct has its place, but it's also a recipe for just driving on faith.

When the Lean Startup came on the scene, a big message there was this idea of **thinking like a scientist, thinking of your project, your startup, your product as an experiment**, and so we can run and learn and use facts to make course corrections along the way.

All that made sense. But there is the dark side, which is if **we go to the other extreme and run a business like scientists, then we would take forever to launch anything because we wouldn't want to build the most perfect experiments.**

If we truly are trying to uncover truths, we should be doing double-blind tests and have no selection bias. We should be looking for what is really true out there.

Business is different in that we want to not uncover perpetual truths, unlike science, we want to find temporal truths that may not even be absolute truths.

These might be quirks of demographics. Maybe the millennials behave a certain way that all of a sudden make certain products possible, and **entrepreneurs are opportunistic.** They need to act on that.

Being able to see signals in the noise and build something that can make the business model work is the goal of the entrepreneur.

Employing the scientific method to run some experiments to get there is helpful, but not at the expense of going overboard and taking too long to find something that will hold true forever, which just doesn't happen in business. **All business models get disrupted.**

What are the key ingredients for an effective entrepreneurial experiment?

First I still think something that is common in both (science and entrepreneurship) is this idea of starting with models. You find a lot of entrepreneurs who kind of latch onto experimentation. What you find is that they just run lots and lots of experiments, but they pick and choose.

They run the experiments that are the most fun, like doing landing page tests or changing words here and there on the copy, which are nice. They're optimization experiments, but they may not be the most critical things in the business.

Where we stand or where I stand is trying to,

Like a scientist, build some models, and for an entrepreneur that is the business model.

Build up a model, use that model to make certain predictions, prioritize what might be riskiest, which is more critical for an entrepreneur than a scientist because they are trying to, in a given timeframe, get to a point where their business can survive.

Finding a business model that works before they run out of resources.

Now to be able to do that we break it down as a model, prioritize, test.

Start with a model, a business model, prioritize what might be riskiest in that model, not what's easiest, and do the hard stuff.

That's the run experiments around those risky assumptions, not necessarily what might just be volumes of experiments.

It's not about the volume but the quality that fundamentally matters.

Why monetization is a building block you want to test early on?

Ash Maurya: It's the way I like to make that difference clear, there's a difference between a hobby and a business as one makes revenue. One has monetization potential or is monetizable and the other isn't.

If the goal here is to build a business model that works, one of the key aspects is ensuring that that business model captures value, in other words, makes money. The sooner we do that the better.

We have from the dotcom days, so even now you see many people deferring that question to later, who is the customer, the customer is the one that buys, but who is that customer and what are they going to pay?

We tend to defer that to later because I find that a lot of entrepreneurs behave like artists. We try to create art, but we don't want to put a price on it because in our minds it's uncomfortable, it devalues that art, we don't sometimes underprice our stuff.

All these biases come into play, but one needs to outgrow those.

If you're really trying to build a business model, that's one of the first things to test. I ask two questions:

- who is the customer? Because in some scenarios you may have some people that are users, others that are customers. Knowing that distinction is very, very important.
- How do you collect money? What is that pricing model? How do you collect it? How much is it? What is that?

So that's that. That would be the second question.

Gennaro Cuofano: I think it's very important to emphasize that if you're creating a company today, it's very important that you try to monetize it very quickly. As you explain also in Scaling Lean, many believe that companies like Facebook didn't make money for a while. But as you explained in the book Facebook actually made money early on. Indeed, Facebook used an approach that you call staged roll out.

What can we learn from the early Facebook on monetization and execution?

Ash Maurya: It's actually a strategy of not launching your product to everyone all at once. We find many examples in the world. I'll share to Facebook being one of them, but Facebook was actually an accidentally staged rollout strategy.

It was accidental because, I don't necessarily know, of course, because I've not spoken to Mark Zuckerberg about this, but through interviews, he was not able to launch to everyone because he was a college student.

They didn't really have the resources to do so. They didn't have the credibility to go and raise money because he wasn't the first social network back then. There were many others out there with millions of dollars in revenue and millions of users as well.

Friendster, Myspace, they were all existing at the time. He couldn't go in and do a public launch, so he launched on one college campus at the time.

The product immediately took off. It had amazing traction on one campus, and that made him pay attention to it.

Of course, he could benchmark his traction versus his closest competitors and saw that this was outperforming them in terms of usage and engagement and monetization potential because usage engagement in that world translates to advertising revenue.

He tested that by putting some Google ads, which was an easy way to test and he was able to show that this was producing twice as much revenue, potentially from a very small number of users, but very highly engaged.

That was the constraint he had is he didn't have money to launch publicly, but by launching on one campus, he could show that this was working quite well.

Now to prove to himself and his team and potential future investors, he went to other campuses and he carefully picked which ones. He went to other schools that have social networks.

Those were his early adopters and his job was to displace the incumbent in there. He also picked Ivy League schools and well-named schools, and so that created also a sense of exclusivity.

By the time they were done with the next four schools, word got out that Facebook is the coolest college social networking platform out there. Only the elite schools have it, and so every other school in the U.S. and maybe other parts of the world wanted to get on that launch list.

They kind of used that very, very strategically to **take a constraint, which is "I don't have resources to publicly launch" and turn it into something where they got a pipeline of schools just waiting to be the next one.**

I can see them going into their investor's office and saying, "You pick the next school for us, and we can demonstrate amazing traction in that school just within 30 days," because they had figured out the system for doing so.

This is what I call in the book a customer factory. They pretty much knew how to go in and get this factory up and running very quickly, where users would come, be highly engaged, they would monetize it, and make the whole business model work.

That's kind of a superpower they gained, which is why behind closed doors investors were getting them huge valuation because they could see their metrics and in these microcosms of markets, which is these different schools.

They could see them lighting them up kind of at, well, they were like little Petri dishes *(a cylindrical glass or plastic lidded dish used for experiments in laboratories)* of experiments that they were running, but because they have repeatability, they could tell a very compelling business model scalability story to their investors, which is why their valuation went through the roof.

How did Tesla instead execute a staged rollout?

Tesla and Elon Musk, the way he launched his electric car also followed a staged rollout. Those of you that know the story even at a high level know that they were three or four cars, but there are three cars that were part of his launch plan, **which was starting with an expensive sports car, the roadster, which was a way for them to test the riskiest assumption**, which was all about the battery.

Can we actually build a new electric battery that can go 200 miles on one charge? They started with that.

That was their stage one. They then went to stage two, which they built that model last, it was a car they built on their own. They had to get good at on building cars. It was all about addressing those risks, and then they finally launched their stage three, which was the less expensive, more affordable electric car. That was the model three.

This was a staged rollout as well. Had Elon Musk gone to stage three in the beginning, which he had the means of doing so. He could've taken the battery that they invented, put that in a cheaper vehicle, and sold that, but he didn't do it because that would have brought on too many risks all at once. He too employed a deliberately staged rollout strategy.

I sometimes like to call that permission to scale, so whether you're a startup, whether you're a corporate, we have to realize that products go through a life cycle, and rather than trying to rush to get to scale prematurely and then making a lot of mistakes and getting overwhelmed with risk, if we instead give ourselves permission to scale, start with smaller numbers of customers and users and more systematically roll it out, it makes the process more manageable. It helps us tackle the riskiest assumptions first.

Gennaro Cuofano: So a staged rollout is a way to go from problem/solution fit to product/market fit, to scale. Isn't it?

How do you go from problem/solution fit to scale?

If I put some goals around it, problem solution fit is proving that we have a big enough market. This is where we search for early adopters. In the roadster case with Tesla, those were those typically tech entrepreneurs or tech workers who are interested in an electric car. Those were the early adopters.

A lot of Silicon Valley and places around there were the early adopters of that vehicle. In the Facebook case, that would be college students on Harvard and those Ivy League schools.

You start with that smaller group and then as you begin to roll out, the product gets to more and more mainstream. That's would be product-market fit, and then that's when we really try to grow well beyond that. That would be the scaling stage.

When is the right time, if there's any, to take venture capital money?

Ash Maurya: I would say in an ideal world, and it's not just me, even the venture capitals and other angel investors will give you the same advice. The best ideal time to raise your big round of funding would be as you cross product-market fit.

Once you have product-market fit, some success is guaranteed. The question is how big can it get?

The other thing that's also going in your favor then is that your goal, you being the entrepreneur or innovator, your goal and the investor's goals are completely aligned. It's all about growth.

You have figured out the product, you have figured out the customer, now it's really engines of growth.

How can we pump in money if you're using paid engines or viral engines? How do we invest in those engines of growth and maximize the potential?

That's something that investors care about, you also care about and so those goals are aligned.

That's the ideal. Now in a realistic sense, everyone has the J curve, so before you get to profitability and product market fit, we have to invest in a product which takes time, money, effort.

When does it make sense to bootstrap?

This is, of course, going to be a function of the kind of business model and product that you have. If you can bootstrap, if you can go all the way to product market fit and then raise money, you maintain the most control in your company. You have a lot of say in where the company goes from that point on.

That's a very powerful place to be, but it's not the case for every kind of product. If you were doing something that required capital investment upfront, this is where VC makes sense, but there is now, increasingly so, a very mature market of other investors that play in that space.

Those would be super angels or angel investors that understand that you still haven't reached product-market fit and you are going to be learning and pivoting and course correcting, and they tend to be more patient for those types of things.

They are taking bigger risks, so they do ask for bigger returns. They give you smaller amounts of money but want a bigger proportional return than a VC would. That's just the nature of investing.

Even earlier than that, if you haven't even launched, it's going to be hard to find even Angel Investors. This is where things like bootstrapping come into play, or the three Fs (friends, families, and fools), kind of a joke that's out there, but that's where raising some seed capital is a way that people get started, and that, of course, tends to be smaller amounts of money just to be able to get that early validation, early traction.

How do you build the optimal business model?

Ash Maurya: This is both the challenge and the opportunity, which is just the nature of the early stage, is that we are trying to find optimal business models, but we can't see the future. We don't know what we don't know. This is where a little bit of that scientific thinking comes in, and we need to experiment.

Even at the business model level, one of the things we are big advocates of takes the idea that you have but create multiple variants, multiple possibilities of places the idea might go.

I might take the same idea and sell it to businesses, so that would be a B2B type of a business model, or it could be B2C, or it could be something different. Maybe it's a marketplace.

We encourage people in the early days to cast a wider net and try to create these multiple variants on the canvas. Go narrow and specific, but you can create a number of canvases.

They can see your idea from many different perspectives, and then we start running some experiments. What we are trying to do is see which of those canvases, which of those business models, which of those customers and markets are reacting the way we think they should, which ones are creating more traction than others? If that aligns with your mission, with your vision, you double down on that.

The reason I say that last part is that sometimes it is possible when we do this kind of searching to find yourself in business models that can work, but maybe don't align with your values.

I have been in that situation a few times. I was trying to build products for entrepreneurs, and I found, for instance, that marketers were getting more excited with the product, and sure, I could go to marketers, and not that I do not like marketers, but it would have gone against a lot of the other assets we had in the company.

We had a brand, we had channels, we'd have to reestablish everything, potentially even create a wholly separate company or identity to be able to go into that customer segment.

We decided not to go there.

It's a hard thing to say no to a good customer problem and potential solution that you have, but sometimes those are things that one has to do.

When you're doing that casting of the canvases of the wider net, it is possible that you will find some promising ideas, some that you want to go down, others that you don't.

You also find a lot of ideas that may seem promising on paper, but when you do a little bit of validation, you find that there may already be existing alternatives. Maybe the customers don't have these problems and so you get rid of them. That generally is the process.

How is traction different from revenue and why is it a key metric?

Ash Maurya: that was the big message in this book (Scaling Lean). The first book was a lot about validation techniques. As we've described already using a canvas to better see your idea and go and do some validation, the challenge, of course, became when you came back and either look for stakeholders or investors, they wanted to see more.

They wanted to see growth potential. They wanted to see things like revenue and profit, and then they, of course, knew that in the early stages of revenue and profit are generally nonexistent because you have to first make that business model work.

What could we use instead? This is where traction comes in.

The key difference between traction and revenue and profit is that the right traction metric is a leading indicator for future revenue and profit.

It's very powerful that can almost predict that revenue and profit will take care of themselves. That's where the power comes in. In the early stages when you don't yet have the revenue and profit to show, if you can show traction instead, that can be a good stand-in.

Facebook in the early days was getting lots of traction. They were getting lots of users hitting their site. They were engaging, they were doing all kinds of things, and so they asked themselves a question is that "This engagement is very high. What if we put ads, would that engagement drop? Would it stick? Would they click through?"

They ran some experiments using Google ads and found that it really did stick.

Today even the way they actually position their story to investors is they talk about the user growth, not so much the advertising growth side because one will control the other.

If you get more users on your platform, engaging more, even daily active users going up, the advertising naturally takes care of itself.

That's the power of thinking in terms of traction versus revenue. One is a leading indicator, while the other is a trailing indicator.

**Note: a leading indicator is input oriented. In short, it is a key driver that determines also a trailing or lagging indicator to move. However, a leading indicator might usually be hard to measure and easy to influence. Contrary to trailing or lagging indicators might be simple to measure but hard to influence.*

What is throughput accounting, and how it works?

Ash Maurya: throughput accounting, it was really a concept I borrowed from Goldratt's work. Those of you that are familiar with the Theory of Constraints, or his popular book, The Goal, that's where he shared this way of thinking, and he applied it in the manufacturing context.

He was working with factories and really got them thinking about throughput. If you're on a factory floor, many people start going into cost-cutting mode. But the problem with cost-cutting is that there is a floor beyond which you can no longer go lower, you can only cut costs so much and then what are you going to do?

But if we flip this around and look at revenue potential, or the potential to make more money, that's almost infinite, at least in theory.

The idea of throughput accounting is flipping that on its head and saying, "Yes, we want to worry about cost-cutting and efficiency, but the bigger potential here is thinking about upside potential."

That's some of the ideas that I share in the book is how to think about throughput accounting in that context in a business modeling context.

Similarly, on a business modeling side, we have the way we deliver value, so that would be the solution you build. Yes, we can make it as efficient as possible in the early days, but that is again chasing pennies and letting dollars slip through the cracks.

What we instead should be doing is focusing more on the revenue streamside, trying to maximize things like pricing, for instance. Trying to identify the right customers, for instance.

Let's work on those problems first, and then we can optimize the cost structure side farther down the road.

How does the 10x growth model work?

Ash Maurya: when we looked at Facebook, for instance, they out of constraints of no money had to go to one campus and then they went to three other campuses and then they kind of rolled out systematically.

10x kind of puts that into more of a systematic context. What we generally tell people, if you're giving yourself permission to scale, instead of thinking of, "I'm not going to launch to everyone," start with one customer.

I know one sounds just wrong, but that is what I call the singularity moment of a product. The moment you can get one person to buy your product and part with their hard earned money, that's something to celebrate.

Now one, of course, is not enough, but if you think about it, every company, whether it's Amazon, Facebook, your company, starts with one customer. Everyone starts in the same place, and then they double and then double subsequently many many times.

If you think of 10x, that is really a sequence of doublings through the power of three is 8x. If you keep doubling, you will eventually 10x, and most people will have 10 customers. You'll need 10x once, some of you will get a hundred customers, you may only need 10x again.

The only difference between a company like Facebook and say, my company or your company, is Facebook just doubles more times in rapid succession and we will, and they keep doubling and then they eventually slow down.

We may not need to get that big because our business model starts working for our scale much, much sooner. The whole idea of 10x-ing is almost giving you a mathematical way of thinking of permission to scale.

The way I break it down is I get, no matter what the idea is, I will get a startup or corporate innovator to start thinking of, "How do I convert my first customer, and then how do I get to 10, how do I get to 100, how do I get to 1,000?"

That nonlinear thinking automatically works towards prioritizing the right types of risks.

If we think about the world we live in today, most products, not all, but most products don't suffer from technical risks. They suffer from customer and market risk. When you are only serving 10 customers, you can fairly easily do that from a technical perspective.

You may only need one web server, for instance, or you may be able to provide high touch onboarding. That allows you to supplement the shortcomings of a fully scalable product.

By giving yourself that permission of doing 10 customers initially, you can do that. As you go to the next level, it's not about getting another 10 customers, you have to get 100 customers. That forces you to start investing incrementally in things that will need to scale.

You don't have to go all the way to scale, but it's an incremental way of not just doing easy jumps. It's still hard, but they are manageable harder steps in that journey.

Why do we need to remove failure from our vocabulary?

Ash Maurya: that's one of the key mindsets that we try to teach entrepreneurs that work with us very, very early on is removing failure from the vocabulary.

Not because you're not going to see it, but the problem that we have, even though we talk a lot about fail fast and failing is the way to learn, all of those things.

At the end of the day, we humans just do not like failure. No one goes out saying, "I failed 10 times today and I'm going to keep doing it." It just doesn't sound very good.

What I've observed, and I've seen this even of myself, is that when we encounter an experiment that has failed, our initial reaction is to try to hide it or to justify it. That's where biases set in.

That's where we need to explain it in a way, and so we throw an explanation which is biased and oftentimes completely wrong to make ourselves feel good. Then we start selling that story to everyone else.

That is moving away from that scientific way of thinking more into covering up or using faith as a way to just explain things away when that isn't really baked in and hide it. I find that "fail fast" is more easily said than done.

In the book, I actually put a quote from Buckminster Fuller who is a celebrated scientist. What he said was:

There is no such thing as a failed experiment, only unexpected outcomes.

The way that I interpret that, is that going back to the whole way of running experiments, we start with a model.

We start with a business model. We make some predictions about how customers will behave, and if they don't behave that way, that's an unexpected outcome.

That's actually something that should be studied because what it's telling you is that your way of thinking was off by some assumption. If you can find that assumption and fix it, you might actually find a breakthrough.

If you actually go to the dictionary definition of breakthrough, there is no breakthrough without some unexpected outcome, because if everything that you expected to happen does happen, then where's the breakthrough?

Easy to rationalize, but I find that that's an opportunity. A breakthrough comes from those unexpected outcomes, calling it failure just makes people act very silly and hide it. But if we instead try to act more like scientists and go after those unexpected outcomes, we may find those breakthrough insights hidden in them.

For me, when I run experiments or when we get other teams to run experiments when they make predictions and they don't come true, we don't just let them sugarcoat it with some justification, we actually want to go deeper.

Call that customer, find out why they didn't buy, or let's run another experiment to see if we can understand some of these root causes. When we have done that, we have found in those answers, you literally find the gems. That's where you find those breakthrough insights. That's where the infection points on the hockey stick curve are hiding.

Key takeaways

- The Internet has enabled an era where we are extremely connected. This brought us from a model of continuous improvement, where continuous testing, experimentation, and learning is the key.
- Entrepreneurship and science are two separate domains. While science does provide a valuable framework for entrepreneurs to borrow. On the other hand, scientists want to uncover universal or more rigorous truths. Entrepreneurs instead look for opportunities, and to achieve business objectives.
- An entrepreneur job is primarily to build a business model that works, by looking at signals in the noise and focusing on the opportunities at hands.

- Making money is one of the keys and riskiest building blocks in a business model. Today, startups tend to think monetization can be deferred as long as possible. But in reality, monetization should be one of the building blocks to test early on, as among the riskiest.
- Constraints, if used strategically, can be used as advantages to spur growth.

- Permission to scale can be used as a way to grow a business by reducing risks and mistakes. Indeed, all products go through a life cycle, and rather than trying to rush to get to scale prematurely, permission to scale enables them to get there gradually.
- The best ideal time to raise your big round of funding would be as you cross product-market fit. At that stage money and speed become critical for the next stage of scale.

- Early on when you didn't figure out yet a viable business model it is important to cast a wider net of variations of business models to test them out, also in parallel.
- A business model is also a matter of choice, based on your entrepreneurial values. Thus, when casting the potential business model to test, you might want to take off those that do not make sense for you as they don't align with your mission.

- Traction should be the key metric for growth. Indeed where traction is a leading indicator, revenue is a trailing indicator. In short, if you focus on traction you also influence revenue.
- A 10x staged rollout enables companies to gain permission to scale, by growing non-linearly, while getting ready to scale.

And a statement worth remembering comes from Buckminster Fuller:

There is no such thing as a failed experiment, only unexpected outcomes.